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Regional Transportation District

RTD Denver EMU

Project # 7230

Door System

Software Project Management Plan (SPMP)

Contract Number	-	CDRL Number	-
		Customer Ref. Number	-

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PREFACE

This document complies with the format and content requirements of the IEEE Std.1058-1998, IEEE Std.1558-2004, and EN50128 standards.

This document is submitted to fulfill the requirements Passenger Entrance Door supplied for DENVER Regional Transportation District (Denver RTD) for the EAGLE P3 project.

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Table of Contents

REVISION LIST **2**

PREFACE **2**

TABLE OF CONTENTS **3**

LIST OF FIGURES **5**

LIST OF TABLES **5**

1 OVERVIEW **6**

1.1 Project Summary 6
1.1.1 Purpose, Scope, and Objectives 6
1.1.2 Assumptions and Constraints 9
1.1.3 Project Deliverables 9
1.1.4 Schedule and Budget Summary 11
1.2 Evolution of the SPMP 11

2 REFERENCES **12**

Standards 12

3 DEFINITIONS **14**

3.1 Acronyms 14

4 PROJECT ORGANIZATION **15**

4.1 External Interfaces 15
4.2 Internal Structure 16
4.3 Roles and Responsibilities 17

5 MANAGERIAL PROCESS PLANS **18**

5.1 Project Start-up Plan 18
5.1.1 Estimation Plan 18
5.1.2 Staffing Plan 18
5.1.3 Resource Acquisition Plan 19
5.1.4 Project Staff Training Plan 19
5.2 Work Plan 20
5.2.1 Work Activities 20
5.2.2 Schedule Allocation 20
5.2.3 Resource Allocation 20
5.2.4 Budget Allocation 20
5.3 Control Plan 20
5.3.1 Requirements Control Plan 20
5.3.2 Schedule Control Plan 20
5.3.3 Budget Control Plan 20
5.3.4 Quality Control Plan 21
5.3.5 Reporting Plan 21
5.3.6 Metrics Collection Plan 23
5.4 Risk Management Plan 24
5.5 Project Closeout Plan 24

6 TECHNICAL PROCESS PLANS **25**

6.1 Process Model 25
6.2 Methods, Tools, and Techniques 25

6.3 Infrastructure Plan 26
6.3.1 Office Space 26
6.3.2 Personal Computers 26
6.4 Product Acceptance Plan 26

7 SUPPORTING PROCESS PLANS 27

7.1 Configuration Management Plan 27
7.2 Verification and Validation Plan 27
7.3 Documentation Plan 27
7.4 Quality Assurance Plan 27
7.5 Reviews and Audits Plan 27
7.6 Problem Resolution Plan 27
7.7 Subcontractor Management Plans 27
7.8 Process Improvement Plan 27

8 ADDITIONAL PLANS 29

APPENDIX A: SOFTWARE PROJECT SCHEDULES 30

APPENDIX B: WORK BREAKDOWN STRUCTURE 31

List of Figures

Figure 1-1: Project Structure 7
Figure 4-1: Software Project Organization - External Interfaces 15
Figure 4-2: Software Project Organization - Internal Structure 16
Figure 6-1: Software Development Process Model 25

List of Tables

Table 1-1: SCI Nomenclature 6
Table 1-2: Software Document Deliverables - CDRL 9
Table 1-3: Deliverables to Production 10
Table 1-4: Budget and Schedule Summary 11
Table 4-1: Responsibilities 17
Table 5-1: Staff Level Needs by Function and by Phase (Persons) 18
Table 6-1: Computers and Software Needs 26

1 OVERVIEW

This document defines the development plan for the Door System of the RTD Denver EMU.

1.1 Project Summary

1.1.1 Purpose, Scope, and Objectives

The purpose of this SPMP is to define software project management responsibilities, resources, processes, and standards, to be followed for all software activities necessary to accomplish the development and maintenance of software for the Door System.

This SPMP and its companion documents, the Software Configuration Management Plan (SCMP), the Software Quality Assurance Plan (SQAP), and the Software Verification & Validation Plan (SVVP) serve as the guiding documents to develop the Door System software project.

This plan is indented for VSRS.'s software project management, the development team, the maintenance team and Hyundai Rotem.

The Door System is composed of a number of subsystems. Table 1-1 identifies the SCI that this SDP applies to. The Figure 1-1 identifies the supporting processes and the SCIs within each subsystem and highlights those that this document applies to.

Table 1-1: SCI Nomenclature

SCI Part Number	SCI Nomenclature	SIL Level
2011L0051	DCU Application	2
2011L0053	DCU Boot Loader	0
2011L0052	Neuron Gateway Application	0
2011L0059	DCU CPLD Binary	2
2011L0054	PTU Application	0

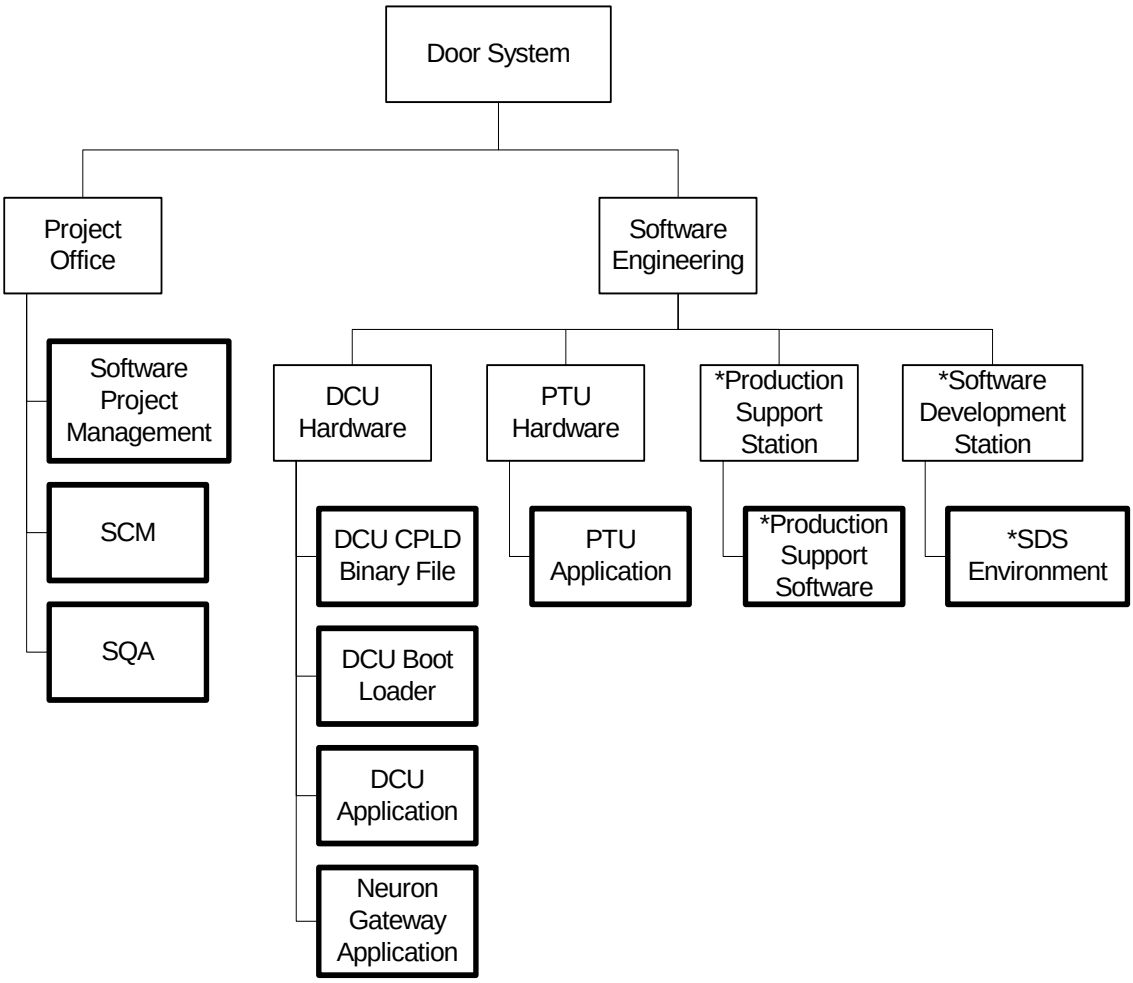


Figure 1-1: Project Structure

*Items are not SCIs/deliverable for this project and will be used internally. (see Sections a), and b) for more description)

The following sub-sections provide an overview of the system.

The PSS hosts the Production Support software.

The SDS hosts all that is required to compile and maintain the above software.

The DCU hosts the DCU Application, the DCU Boot Loader, the Neuron Gateway Application, and the DCU CPLD.

The PTU hosts the PTU Application software.

1.1.1.1 VSRS Internal Items

a) Production Support Station (PSS):

This station is for the internal use only, and it is not deliverable. Vapor's production team will use this station for the production.

- The Production Support Software is part of the PSS to control the following functions:
  - Upload the DCU Boot Loader Application and DCU Application into the DCU;
  - Upload the DCU Neuron Gateway Application into the Neuron Gateway processor;
  - Upload the DCU CPLD file into the CPLD chip;
  - Test the hardware using internal test tools, namely the CPU Test and RTP Tool;
  - Interfaces to the test technician, providing commands and test results;
  - Allows monitoring of DCU unit(s) while in the Burn-In environment chambers.

b) Software Development Station (SDS):

This station is for the internal use only, and it is not deliverable. Vapor's software team will use this station for the development of the SCI.

- The Environment is part of the SDS and allows to:
  - Maintain (edit, modify and recompile) all above software;
  - Maintain (edit and print) the documentation of all above software, including attachments and embedded objects, but excluding AutoCAD and Solid-Edge drawings;
  - Maintain (edit, modify and recompile) programmable devices;
  - Create the media (e.g. floppy disks, CDROM) and labels required for release of the maintained items, if any;
  - Upload the DCU Boot Loader Application and DCU Application into the DCU for the Development purposes;
  - Upload the DCU Neuron Gateway Application into the Neuron Gateway processor;
  - Allows programming the CPLD.

1.1.1.2 Deliverable and Configuration Items

a) Door Control Unit (DCU)

The following provides a brief overview of each SCI:

- The DCU Boot Loader is part of the DCU to control the following functions:
  - Initial CPU and peripheral initialization
  - Initial self test and memory integrity checks
  - DCU Application Update

- The DCU Application is part of the DCU to control the following functions:
  - Door Cycling
  - Audible & Visual Warning Indications
  - Monitoring Operations
  - Data Logging Operations
  - Diagnostics
  - Troubleshooting
  - PTU Communication Interface (host side)
  - Interface to the trainlines and the Door Control Panel
  - Interface with the SCI Neuron Application

- The Neuron Gateway Application is part of the DCU to control the following functions:
  - Handles the LonWorks network protocol
  - Interfaces to the DCU Application
  - Interfaces to the Car Network.

- The DCU CPLD Binary File is part of the DCU to control internal logic. Note that the Electrical team is responsible for generating the CPLD Binary file.

b) Portable Test Unit (PTU)

- The PTU Application software is part of the PTU to control the following functions:
  - Software upload into the DCU
  - Monitoring
  - Data logging retrieval
  - Troubleshooting

1.1.2 Assumptions and Constraints

The following lists the factors that limit the software developer's option:

- The software shall be designed such that its implementation fits within the resources available on its specific hardware;
- The PTU Application shall be designed to work on the Microsoft Windows XP™, Windows VISTA™ and Windows 7™ operating systems;
- Software development shall comply with the Vapor's Software Coding Standard document #SWI-02. The purpose of this coding standard is to improve maintainability, reusability, and source-code quality;
- Software development shall comply with the EN50128 standard;
- Software documentation shall comply with IEEE standards and VSRS internal procedures.

1.1.3 Project Deliverables

Table 1-2 lists the work products that will be delivered to the customer, the CDRL number, the quantity, the media, and the delivery date.

Please note that the *Safety Related Documents* in Table 1-2 are produced and maintained by the RAMS engineering group to comply with the safety requirement of the VSRS safety procedure and the EN50128 standards. The Software Quality Assurance Plan (SQAP) 201111208, Section 3.1 Organization, explains the role and the task of the RAMS engineer.

Table 1-3 lists the work products that will be delivered to the production, the quantity, the media (or form), and the delivery date.

Table 1-2: Software Document Deliverables - CDRL

VSRS Part #	Document Name	Date
	Safety Related Documents	
201111162	Reliability, Maintainability, and Safety (RM&S) Program Plan	Pre-PDR
201111340	Preliminary Hazard Analysis (PHA)	Pre-PDR
TBD	Hazard Log	Pre-PDR
TBD	Safety Requirements Specification	Post-PDR
TBD	Safety Validation Plan	Post-PDR
TBD	Failure Mode Effects and Criticality Analysis (FMECA)	FDR
TBD	Fault Tree Analysis (FTA)	FDR
TBD	Operating and Support Hazard Analysis (O&SHA)	FDR
TBD	Sub-System Hazard Analysis (SSHA)	FDR
TBD	System Hazard Analysis (SHA)	FDR
TBD	Flame Smoke and Toxicity Analysis	FDR

VSRS Part #	Document Name	Date
TBD	Reliability Prediction	FDR
TBD	Service Failure Predictions	FDR
TBD	Software Assessment Report	FDR
TBD	Safety Validation Report	FDR
TBD	Safety Verification Report	FDR
	System Documents	
201111186	SCMP	Pre-PDR
201111207	SPMP	Post-PDR
201111208	SQAP	Pre-PDR
201111185	SFD	Pre-PDR
201111254	SVVP	Post-PDR
201111446	SVVR	FAI
201111447	SCIST	FDR
201111448	SRTM	FDR
201111449	SVD	FAI
	DCU Application Documents	
201111255	SRS	FDR
201111450	SDD	FDR
201111451	SCTP*	FDR
201111452	SCTR	FAI
201111453	SITP**	FDR
201111454	SITR	FAI
	DCU Boot Loader Application Documents	
201111455	SRS	FDR
201111456	SDD	FDR
201111457	SITP**	FDR
201111458	SITR	FAI
	Neuron Gateway Application Documents	
201111256	SRS	FDR
201111459	SDD	FDR
201111460	SITP**	FDR
201111461	SITR	FAI
	PTU Application Documents	
201111462	SRS	FDR
201111463	SDD	FDR
201111464	SITP**	FDR
201111465	SITR	FAI
201111466	SUD	FDR

* As documented in the SQAP #2011208 Section 4.9 Software Test Plans, SCTP will include Software Component Test Planning, Software Component Test Design Specification, and Software Component Test Procedures Specification.
** As documented in the SQAP #2011208 Section 4.9 Software Test Plans, SITP will include Software Integration Test Planning, Software Integration Test Design Specification, and Software Integration Test Procedures Specification.

Table 1-3: Deliverables to Production

#	Description	Qty	Media/Form
2011L0051	DCU Application Executable	1	Zip file
2011L0053	DCU Boot Loader Application Executable	1	Zip file
2011L0052	Neuron Gateway Application Executable	1	Zip file
2011L0054	PTU Application Executable	1	Zip file

#	Description	Qty	Media/Form
201111468	CD-ROM Media with Door System Software (DCU, DCU Boot Loader, and Neuron Gateway Application)	3	CD-ROM
201111469	CD-ROM Media with PTU Application Software	3	CD-ROM
201111470	CD-ROM Media with SW Engineering Documentation	3	CD-ROM
N/A	License(s) for all software to allow Denver usage	1/Software	Paper

1.1.4 Schedule and Budget Summary

The software project schedule is presented in Appendix A. This schedule shows activities and required resources to perform the Denver software project. The Denver Door System software project will follow the process and activities defined in the Software Engineering Process (SEP) CMM-01 tailored as per the minutes of Software Project Review Meeting #3. Note that the schedule presents the budget estimated in hours.

The project activities statuses are verified every two weeks during the SPRMs throughout the complete life cycle of the project.

Table 1-4 below provides a summary of the major steps workload and finish dates for each step. Please note that more precise finish date will be generated in the next revision of this SPMP.

Table 1-4: Budget and Schedule Summary

SEP Modules	Description	Workload (Hours)	Finish Date
SPP-200	Software Project Planning	129	-
SPG-300	Software Plans Generation	189.5	-
SRD-400	Software Requirements Definition	594.02	Pre-PDR
SD-500	Software Development	1390.47	FDR
SCT-600	Software Coding and Testing	805.95	Post-FDR
SV-700	Software Validation	481.72	FAI
SDT-800	Software Development Tracking	285.19	Last Production Unit
	Total Hours	3875.85	

1.2 Evolution of the SPMP

This SPMP shall be maintained and modified to reflect the current plans, policies, processes, resources, and standards affecting the Door System software project. It is the Software Team Leader's responsibility to keep abreast of industry technology changes and programmatic direction from the Project Manager that would require modification of this plan.

Releases of the SPMP are distributed to team members at the SPRM (Software Progress Review Meeting), the PRM (Project Review Meeting), and the SRM (Senior Review Meeting). Refer to the Door System's SCMP document #201111186 for details about documentation release and configuration control processes.

2 REFERENCES

This section lists the number, title, revision, date, and source of all documents referenced in this SPMP.

Standards

CMU/SEI-2006-TR-008	CMMI® for Development, Version 1.2, August 2006, Software Engineering Institute, Carnegie Mellon University, www.sei.cmu.edu
IEEE Std 730-2002	IEEE Standards for Software Quality Assurance Plans, September 23, 2002, IEEE, www.ieee.org
IEEE Std 828-1998	IEEE Standards for Software Configuration Management Plans, June 25, 1998, IEEE, www.ieee.org
IEEE Std 829-1998	IEEE Standards for Software Test Documentation, September 16,1998, IEEE, www.ieee.org
IEEE Std 830-1998	IEEE Recommended Practice for Software Requirements Specification, June 25, 1998, IEEE, www.ieee.org
IEEE Std 1012-1998	IEEE Standards for Software Verification and Validation, March 9, 1998, IEEE, www.ieee.org
IEEE Std 1016-2009	IEEE Recommended Practice for Software Design Description, July 20, 2009, IEEE, www.ieee.org
IEEE Std 1058-1998	IEEE Standard for Software Project Management Plans, December 8, 1998, IEEE, www.ieee.org
IEEE Std 1558-2004	IEEE Standard for Software Documentation for Rail Equipment and Systems, March 23, 2005, IEEE, www.ieee.org
IEEE Std 1063-2001	IEEE Standards for Software User Documentation, December 20, 2001, IEEE, www.ieee.org
IEEE Std 1058-1998	IEEE Standard for Software Project Management Plans, 8 December 1998, IEEE, www.ieee.org
EN50128	European Standard for Railways Application – Communications, Signaling & Processing Systems – Software for Railway Control & Protection Systems, March 2001
CMU/SEI-93-TR-25	Key Practices of the Capability Maturity Model, Version 1.1, February 1993, Software Engineering Institute, Carnegie Mellon University, www.sei.cmu.edu

VSRS Documents

CMM-01	Software Engineering Process, Rev.3, 2001-03-31
CMMI-02	Software Engineering Policy Manual, Rev.1, 2008-07-18
SWI-02	Coding Standard Guide, Rev.3, 2001-03-27
SWI-03	Software Estimation Guide, Rev.3, 2000-08-23
SWI-05	Inspection Guide, Rev.4, 2003-02-20
SWI-06	Code Walkthrough Guide, Rev.1, 2000-03-03
SWI-07	Lessons Learned Guide, Rev.1, 2000-03-03
SWI-08	SPRM Guide, Rev.2, 2000-03-03

SWI-10	SRM Guide, Rev.3, 2001-03-31
SWI-16	Risk Guide, Rev.2, 2000-03-03
S01-01	Software Problem Report Form, Rev.21, 2001-03-30
S03-01	Estimation Template, Rev.3, 2000-11-07
S05-01	Inspection Form (Database), Rev.4.2, 2000-01-30
S05-02 to S05-15	Documentation Inspection Checklist Template, Rev.1, 2003-02-20
S06-01	Code Walkthrough Form (Database), Rev. 4.2, 2001-01-30
S07-01	Lessons Learn Form, Rev.1.2, 2000-08-21
S08-01	SPRM MoM, Rev.3, 1999-12-08
S10-01	SRM MoM, Rev.2.2, 2000-06-22
S13-01	Audit Notification, Rev.2, 2003-04-01
S13-02	Improvement Action Request Form, Rev.1.1, 2001-01-30
S13-03 to S13-21	Audit Checklist Template, Rev.1, 2003-05-29
S13-22	Audit Report Template, Rev.1, 2003-04-01
S16-01	Risk Form, Rev.2.1, 1996-10-14
S17-02	Software Project Management Plan Template, Rev.1, 2003-01-16
201111185	System Functional Description, Door System for Regional Transportation District, Rev. 0, Date Unknown
201111208	Software Quality Assurance Plan, Door System for Regional Transportation District, Rev. 1, 2011-05-31
201111186	Software Configuration Management Plan, Door System for Regional Transportation District, Rev. 1, 2011-06-02
201111254	Software Verification and Validation Plan, Door System for Regional Transportation District – DCU Application, Rev. 0, Date Unknown
SPRM #3	Software Project Review Meeting Minutes of meeting #3, 2011-05-20

Hyundai Rotem Documents

REDE112784	RTD Denver EMU (50) Procurement Technical Specification of Entrance door System for Denver Eagle P3 Project
REDE113777	Function Description for Side Door System Denver EMU (50Cars)

RTD Denver Document

RFP No. 18FH012	RTD FasTracks Request for Proposal, 2009-09-30

Other Documents

ISBN-10: 0534033830	Project Management for Engineers, Milton D. Rosenau, August, 1984

3 DEFINITIONS

The definitions and acronyms listed below establish meaning in the context of this SPMP.

Software Configuration Item: An aggregation of software that is designated for configuration management and treated as a single entity in the configuration management process.

Configuration Management (CM): A discipline applying technical and administrative direction and surveillance to: identify and document the functional and physical characteristics of a configuration item, control changes to those characteristics, record and report change processing and implementation status, and verify compliance with specified requirements.

Product Baseline: The initial approved technical documentation (including, for software, source code), defining a configuration item during the production, operation, maintenance and logistics support of its life cycle.

Quality Assurance: A planned and systematic pattern of all actions necessary to provide adequate confidence that the item or product conforms to established technical requirements.

Software Project Management: Application of technical and administrative management to a software project.

3.1 Acronyms

ACWP	Actual Cost of Work Performed
BAC	Budgeted cost At Completion
BCWP	Budgeted Cost of Work Performed
BCWS	Budgeted Cost of Work Scheduled
BTE	Bench Test Equipment
CCB	Configuration Control Board
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CDS	Central Diagnostics System
CMM	Capability Maturity Model
CPLD	Complex Programmable Logic Device
CSCI	Computer Software Configuration Item
DCG	Document Control Group
DCU	Door Control Module
DCU	Door Control Unit
EAC	Estimated cost At Completion
ECD	Engineering Change Description
EPROM	Erasable Programmable Read-Only Memory
FCA	Functional Configuration Audit
IAR	Improvement Action Request
IDE	Integrated Development Environment
IEEE	Institute of Electrical and Electronics Engineers
LOC	Lines Of Code
MIS	Management Information System
MoM	Minutes of Meeting
N/A	Not Applicable
OBS	Organizational Breakdown Structure
PCA	Physical Configuration Audit
PDR	Preliminary Design Review
PTU	Portable Interface Unit
PM	Project Manager
PRM	Progress Review Meeting
PSS	Production Support Station
QTP	Qualification Test Procedure
RTM	Requirements Traceability Matrix
RTP	Routine Test Procedure
SCCB	Software Configuration Control Board
SCI	Software Configuration Item
SCIST	Software Configuration Items Summary Table
SCM	Software Configuration Management/Manager
SCMP	SCM Plan
SDD	Software Design Description
SPMP	Software Development Plan
SDS	Software Development Station
SE	System Engineer
SEP	Software Engineering Process
SEPG	Software Engineering Process Group
SLOC	Source Lines of Code
SPM	Software Project Manager
SPR	Software Problem Report
SPRM	Software Project Review Meeting
SQA	Software Quality Assurance
SQAP	Software Quality Assurance Plan
SRM	Senior Review Meeting
SRS	Software Requirements Specification
SRVTM	Software Requirements Verification and Traceability Matrix
SSDD	System/Subsystem Design Description
SSR	Software Specifications Review
SSS	System/Subsystem Specification
SVVP	Software Verification and Validation Plan
SWE	Software Engineer
TBD	To be determined
TRR	Test Readiness Review
VSRS	Vapor Stone Rail Systems .
WBS	Work Breakdown Structure

4 PROJECT ORGANIZATION

This section of the SPMP identifies the interfaces to organizational entities external to the project, the project's internal organizational structure and defines the roles and responsibilities for each.

4.1 External Interfaces

The Figure 4-1 is a graphic representation of the project organization structure with respect to the Door System software project. Although the Software Project Manager takes direction from the Project Manager, it operates within the policies and procedures established by VSRS 's Engineering Manager.

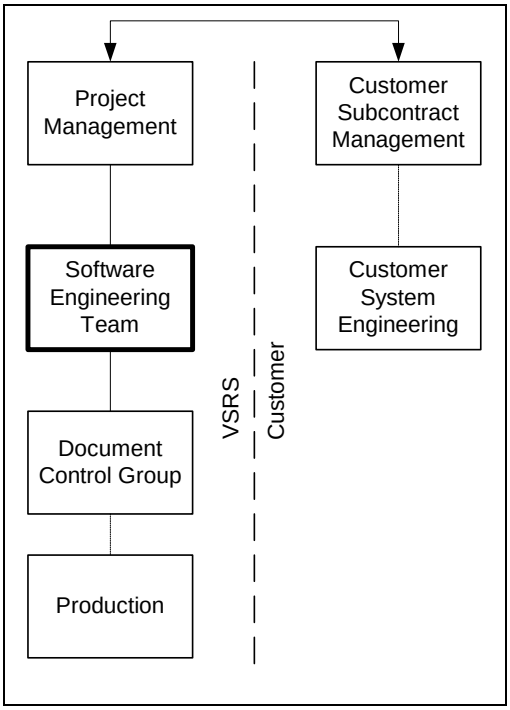


Figure 4-1: Software Project Organization - External Interfaces

Listed below are the responsibilities of each of the organizations and its relation to the software project.

Project Management – Project manager has the authority to ensure complete fulfillment of all project requirements. The Project Manager (PM) has the overall responsibility for acquisition, funding, and transitioning of the project. The PM is the point of contact with the customer. PM also ensures compliance with all contract requirements as defined in the Contract Data Requirements List (CDRL).

Software Engineering Team – Responsible for the software aspects of the project. The software team has the responsibility for local funding, allocations, scheduling, tasking, and reporting to project management. In addition, the team interfaces with the customer in matters of technical nature.

Document Control Group – The Document Control Group (DCG) is responsible for the receipt, distribution, and tracking of technical data associated with the project. The DCG is a service provided by the Engineering department. The DCG releases the work products to the production department under the directives of the projects.

Customer Subcontract Management – The Project Manager interfaces to the Customer's Subcontract Management group for matters of contractual and technical nature. The Customer Subcontract Manager is the point of contact of the project with the customer.

Customer System Engineering – The project interfaces to the Customer's System Engineering group for matters of system interfaces (e.g. Train network interface) through the Customer Subcontract Management.

4.2 Internal Structure

Figure 4-2 is a graphic representation of the software project internal structure with respect to the Door System project. Although the Software Quality Assurance is part of the software project, the SQA takes directions from the Quality Assurance Manager.

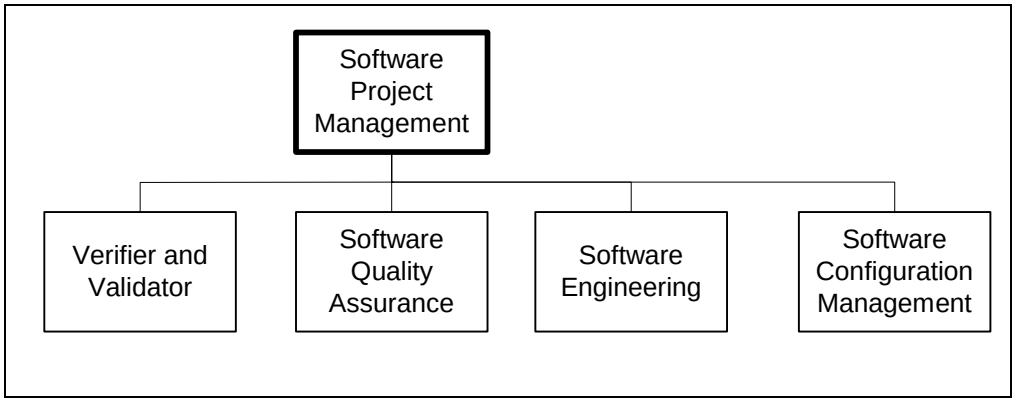


Figure 4-2: Software Project Organization - Internal Structure

Listed below are the responsibilities of each of the software related group and its function within the software project.

Software Project Management – Responsible for the technical aspects of the project. The Software Project Manager (SPM) has the responsibility for local funding, allocations, scheduling, tasking, and reporting to project management. In addition, the SPM interfaces with the customer in matters of technical nature. Responsible for ensuring that changes made to a system are supportable.

Verifier and Validator – As per the Standard EN 50128 SIL level 2 for the DCU Application, Verifier and Validator is responsible for producing test plans, test design, test cases, procedures, reports, testing and verification, and acceptance testing. V&V is responsible for the conduct of SCI tests and incorporates approved changes into released test documentation based on change request baseline data from SCM. SCM identifies all change requests included in an engineering release that should be tested. V&V then provides SCM a copy of the test reports.

Software Quality Assurance – The SQA is responsible for auditing the software development activities and products (FCA and PCA) and certifying of SCM compliance with this plan. The SQA confirms verification of change request corrective measures prior to change request closure. Responsible for administering the verification and validation (V&V) tests prior to release of the software system.

Software Engineering – Responsible for software requirements & design (and associated documentation) overview and guidance; detailed design and coding. Designer ensures that the correct configuration is undergoing the test, and also makes sure that all requirements are traced correctly.

Software Configuration Management – The Software Configuration Manager (SCM) is responsible for the identification, control, status accounting, and supports configuration audits and reviews of the software work products. The SCM releases work products to the DCG with the directives of Software Project Manager.

4.3 Roles and Responsibilities

Table 4-1 identifies the specific project's organizational elements responsibilities (as per Section 4.2) against the activities of Section 5, 6, 7, and 8.

Table 4-1: Responsibilities

Activity as per Section 5, 6, 7, and 8	Responsibility as per Section 4.2
5 Managerial Process Plans	
5.1 Project Start-up Plan	Software Project Manager
5.2 Work Plan	Software Project Manager
5.3 Control Plan	Software Project Manager
5.4 Risk Management Plan	Software Project Manager
5.5 Project Closeout Plan	Software Project Manager
6 Technical Process Plans	Software Project Manager
7 Supporting Process Plans	
7.1 Configuration Management Plan	Software Configuration Manager
7.2 Verification and Validation Plan	Verifier and Validator
7.3 Documentation Plan	Software Engineering
7.4 Quality Assurance Plan	Software Quality Assurance
7.5 Reviews and Audits Plan	Software Project Manager Software Quality Assurance
7.6 Problem Resolution Plan	Software Configuration Manager
7.7 Subcontractor Management Plans	N/A
7.8 Process Improvement Plan	Software Project Manager
8 Additional Plans	RAMS Engineering Group

5 MANAGERIAL PROCESS PLANS

This section of the SPMP specifies the project management processes for the project. This section includes the project start-up plan, risk management plan, project work plan, project control plan, and project closeout plan.

5.1 Project Start-up Plan

This section specifies the estimation plan, staffing plan, resource acquisition plan, and the training plan.

5.1.1 Estimation Plan

The workload presented in the WBS in Appendix B:Work Breakdown Structure is including the following activity breakdown for each generated document:

- QA Overhead: 1 hour
- Configuration Management: 1 hour
- Remaining workload is separated using the following percentage:
  - Document Update: 52%
  - Technical Inspection: 22%
  - QA Inspection: 17%
  - Post-inspection Update 07 %
  - Final acceptance & Signature 02%

The method used for preparing the WBS is based on the method presented in the book “Project Management for Engineer” by Milton D. Rosenau, Jr.

With the same method, the schedule will be prepared using the estimated time which is mathematically based on the optimistic time, the mean time and the pessimistic time.

Re-estimation is to be performed whenever the project schedule performance index (SPI) is going outside of an acceptable range.

5.1.2 Staffing Plan

Table 5-1 specifies the number of staff required by function, and by Door System project phase. The superscripts next to the numbers indicate the source of staff personnel.

Table 5-1: Staff Level Needs by Function and by Phase (Persons)

Staff Function / Project Phase	System Requirements Analysis	System Design	Software Requirements Analysis	Software Design	Software Coding & Unit Tests	Software Validation	System Integration Tests	System Qualification
Software Project Manager	1^T	1	1	1	1	1	1	1
System Engineer	1^T	1	1	1	1	1	1	1
Software Engineering	-	1^T	$1+1^T$	2	2	2	1	1
Verifier & Validator	-	-	1^T	1	1	1	1	1
Software Quality Assurance	1^T	1	1	1	1	1	1	1
Software Configuration Management	1^T	1	1	1	1	1	1	1
Total	4	5	7	7	7	7	6	6

Legend:

N = New Hire
T = Internal Transfer
C = Contracted

5.1.3 Resource Acquisition Plan

All the resources for the project are already available.

5.1.4 Project Staff Training Plan

For training that is provided by the VSRS 's Software Engineering Process Group (SEPG), the training needs are communicated to the SEPG Coordinator to be incorporated in the annual SEPG Training Plan.

The Human Resources department is responsible to provide general administrative training.

The Software Project Manager is responsible for providing training for product-specific and project-specific technical skills.

The Software Configuration Manager is responsible for providing project-specific configuration management procedures training.

The Software Quality Assurance is responsible to provide orientation on the role, responsibilities, authority, and value of the SQA group.

5.2 Work Plan

This section of the SPMP specifies the work activities, schedule, resources, and budget details for the software project.

5.2.1 Work Activities

The Door System project's Work Breakdown Structure is presented in Appendix B. Some activities (e.g. System Engineering) are not directly part of the software project, but are listed since participation of the software team is required.

5.2.2 Schedule Allocation

The Appendix B lists the work activities, their relationships, the estimated start and end dates. The work activities are defined with a clear, unambiguous name depicting a self-explanatory completion criterion (e.g. Draft SRS & inspection sheet).

5.2.3 Resource Allocation

A summary of human resources allocation is documented in Table 5-1. The Appendix B provides a detailed itemization of resource allocation.

5.2.4 Budget Allocation

A summary of budgeted cost allocation is documented in Table 1-4. The Appendix B provides a detailed itemization of budgeted cost allocation.

5.3 Control Plan

This Section of the SPMP specifies the metrics, reporting mechanisms, and control procedures necessary to measure, report, and control the product requirements, the project schedule, budget, and resources, and the quality of work processes and work products.

5.3.1 Requirements Control Plan

Traceability of requirements will be maintained by flowing down from the System Functional Description (SFD) document to the Software Requirements Specifications (SRS) documents, to the Software Design Description (SDD) documents, and finally, to the Software Integration Test Plan (SITP) documents.

Software documentation traceability will be documented in the Software Requirements Traceability Matrix (SRTM) document.

The Rational's RequisitePro tool will be used to maintain traceability of requirements and perform impact analysis upon change requests.

The Software Configuration Control Board and change control are documented in the SCMP document, Section 2.2 and 3.2 respectively.

5.3.2 Schedule Control Plan

Schedule control will be performed as documented in Sections 5.3.5.2.2 and 5.3.6.

5.3.3 Budget Control Plan

Budget control will be performed using standard Earned Value techniques as documented in Sections 5.3.5.2 and 5.3.6.

5.3.4 Quality Control Plan

In addition to the quality controls documented in Section 7.4, and the verification and validation controls documented in Section 7.2, all deliverables listed in Section 1.1.3 will undergo the Inspection Process as per SWI-05. Source code will undergo the Code Walkthrough process as per SWI-06.

5.3.5 Reporting Plan

This Section specifies the reporting mechanisms, report format, and information flows used in communicating the project status and project metrics.

The next sections depicts:

- Reporting Mechanisms
- Reporting Requirements

5.3.5.1 Reporting Mechanisms

Reports are communicated to three levels: the software-related team level, the project management level, and to senior management level.

5.3.5.1.1 Software Project Review Meeting (SPRM)

The software project team will meet once every two to three weeks to perform a Software Project Review Meeting as per the SWI-08. Minutes and action items will be recorded onto an S08-01 form.

Engineering and supporting processes plans, progress, risks, and issues are discussed at the SPRM.

5.3.5.1.2 Senior Review Meeting (SRM)

The software project team will meet the Engineering Manager once every four to six weeks to perform a Senior Review Meeting as per the SWI-10. Minutes and action items will be recorded onto an S10-01 form.

5.3.5.1.3 Project Review Meeting (PRM)

The Software Project Manager attends the Project Manager's Project Review meeting, along with the other project managers.

The Project Manager chairs and schedules this meeting once every four to six weeks, and on an event-basis, where engineering and supporting processes progress, risks, and issues at the project-level are discussed.

5.3.5.2 Reporting Requirements

This section delineates the reports that will be prepared.

The following reports are prepared to assess the project status:

- Earned Value
- Milestone Dates
- Requirements Status
- Problem Report Status
- ECD Status
- Improvement Action Requests Status
- Waiver/Deviation Requests Status
- Product Size

5.3.5.2.1 Earned Value

Earned Value compares the cost of work performed to the budget, based on dollars budgeted for each item in the Work Breakdown Structure. The measure helps to identify cost overruns and under-runs early in the project.

The following will be presented:

- Budgeted cost of work scheduled (BCWS)
- Budgeted cost of work performed (BCWP)
- Actual cost of work performed (ACWP)
- Estimated cost at completion (EAC)
- Budgeted cost at completion (BAC)

Data will be collected to a level of detail in order to have no more than 4 weeks of effort per WBS item. For reporting, data will be aggregated to at least two levels further than presented in Table 1-4.

Data will be presented both numerically and on a time-graph.

5.3.5.2.2 Milestone Dates

The planned vs. actual start or end dates of major project activity will be presented in an MS-Excel worksheet. The following information will be presented for each milestone:

- Activity or event name
- Version of the plan/schedule
- Planned Date (start, end, or both)
- Actual Date (start, end, or both)

The data will be collected for each SCI, for quality assurance, and for configuration management major milestones.

5.3.5.2.3 Requirements Status

Requirements status will present the evolution of:

- Total number of system requirements allocated to each software
- Number of software requirements traced to software specifications
- Number of software requirements traced to software design element
- Number of software requirements traced to test cases and procedures
- Number of software requirements tested successfully

5.3.5.2.4 Problem Report Status

Problem report status counts the number of software problems (or system problem attributed to software) reported and resolved. The rates at which problem reports are written and resolved will provide an indication of product maturity and readiness for delivery.

The following information will be collected and presented for each software:

- Number of problem reports reported
- Number of problem reports resolved
- Average age of problems
- Average time to resolve

The number of problem reports reported and resolved will be presented on a time graph.

5.3.5.2.5 ECD Status

Engineering Change Requests (ECD) status counts the number of changes requests that affect the software system. The measure provides an indication of the amount of rework that has been performed or will be required.

The following information will be collected as a whole for all project's software:

- Number of ECD approved for implementation
- Number of ECD tested
- Number of ECD released

The numbers will be presented both numerically and on a time graph.

5.3.5.2.6 Improvement Action Requests Status

IAR status counts the number of Software Quality Assurance's Improvement Action Requests generated and resolved. The rates at which IAR are reported and resolved will provide an indication of process maturity and readiness to undergo the next project phase.

The following information will be collected and presented for each software:

- Status or each IAR
- Number of IAR reported
- Number of IAR resolved
- Average age of open IAR
- Average time to resolve

The information will be presented both in tabular form and on a time graph.

5.3.5.2.7 Waiver/Deviation Requests Status

Waiver/Deviation status counts the number of waivers or deviations requested, approved, and rejected. The moment and rates at which Waiver/Deviations are reported and resolved will provide an indication on project risks, requirements stability, and approval-authority responsiveness.

The following information will be collected and presented for system's software-related waiver/deviations:

- Status or each Waiver
- Number of waivers requested
- Number of waivers approved
- Number of waivers rejected
- Average age of requested waivers
- Shorted, average, and longest time to get response

The information will be presented both in tabular form and on a time graph.

5.3.5.2.8 Product Size

The Lines of Code (LOC) measure counts the total amount of source code and the amount that has been added, modified, or deleted. Helps in estimating project costs, required effort, schedule, and productivity.

The following information will be aggregated to the software-module level.

- Number of lines of code (LOC)
- Number of LOC added
- Number of LOC deleted
- Number of LOC modified

The report will provide the values in tabular form, and summary to the SCI-level and the project-level. The summary will be also shown as a time-graph.

5.3.6 Metrics Collection Plan

This section specifies the methods, tools, and techniques used in collecting and retaining project metrics. The metrics reporting are documented in Section 5.3.5.

All metrics collection will occur once per month, on the last day of the month.

Detailed metrics collection procedures (documents) will be prepared and maintained to support the reporting documented in Section 5.3.5. The procedures will specify the tool(s) to use, the specific steps to gather metrics data, how & where to record them, and how to generate the required reports. The SQAP document #201111208 Section 5.6 SQA Metrics also discusses the metrics collection plan.

The procedures, maintained as separate documents, are:

- Effort, Estimates, and Earned Value Metrics Procedure

- Milestone Dates Metrics Collection Procedure
- Requirements Metrics Collection Procedure
- Configuration Management Metrics Collection Procedure
- Software Quality Metrics Collection Procedure

5.4 Risk Management Plan

Risk management will be performed as per the SWI-16 using the S16-01 standard form. The Software Project Manager is responsible performing software risk management.

5.5 Project Closeout Plan

This section of the SPMP contains the plans necessary to ensure orderly closeout of the software project.

A project's Lessons Learned session will be performed at project completion of the development phase. The session will proceed as per the SWI-07, and the results will be recorded onto S07-01 forms. All software-related groups will participate in the lessons learned session.

Project staff will be reassigned to other projects but will also form, on a part-time basis, the maintenance staff for the software maintenance part of the project's life cycle.

6 TECHNICAL PROCESS PLANS

This section of the SPMP specifies the development process model, the technical methods, tools, and techniques that are used to develop the various work products.

6.1 Process Model

The Door System software development will conform to the waterfall model. The development approach will apply Level 2 and selected Level 3 software engineering processes in accordance with the Software Engineering Institute (SEI) Capability Maturity Model (CMM), in conformance to the VSRS 's Software Engineering Process (ref. document CMM-01) and policies (ref. document CMM-02).

The Software Development Process Model is shown in Figure 6-1.

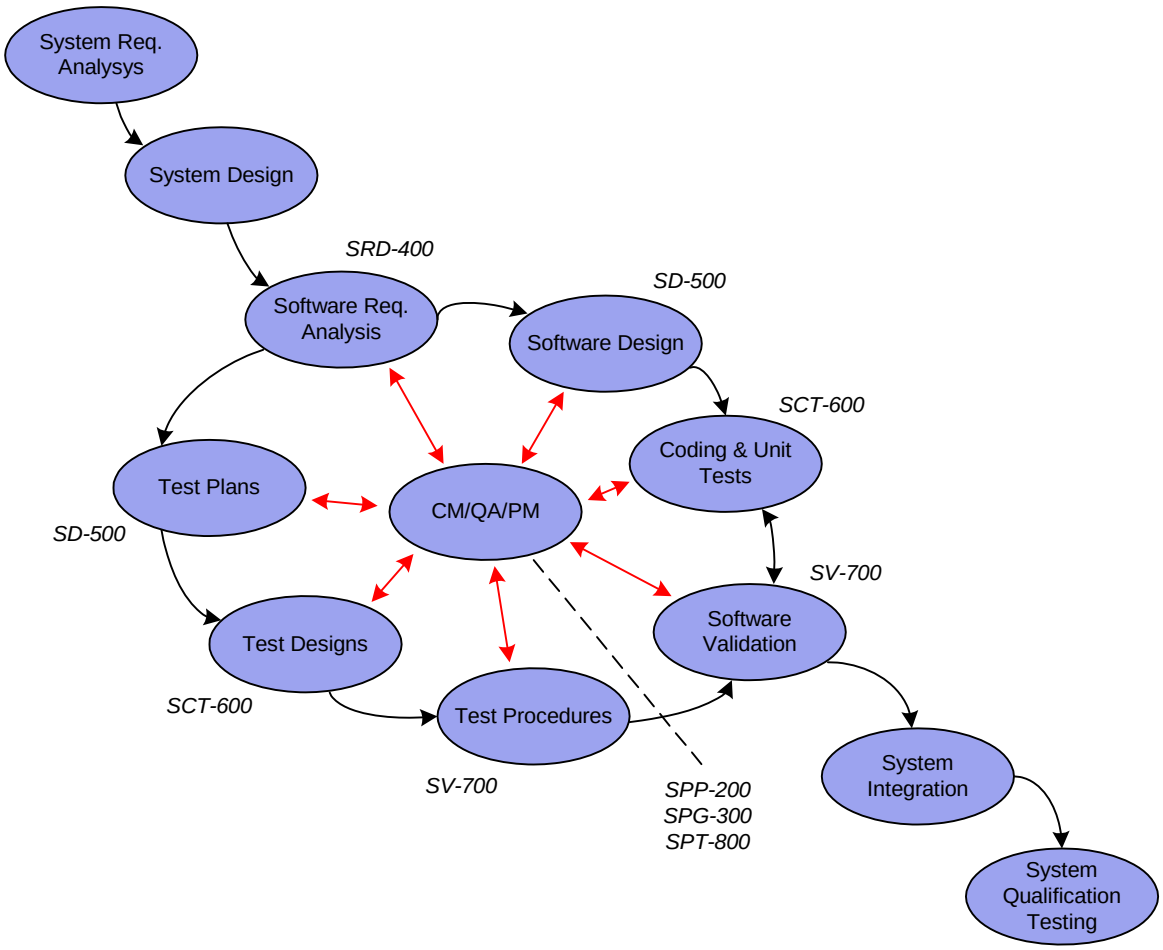


Figure 6-1: Software Development Process Model

6.2 Methods, Tools, and Techniques

Documents will be written using, or be compatible with, Microsoft Office (i.e. Word, Excel, Visio) in conjunction with Rational's RequisitePro for handling traceability.

The design methodology will use functional decomposition as the mean for designing software.

The software development tools & environments are documented in the Door System's Software Configuration Items Summary Table document 201111447.

The VSRS 's SWI-02 governs the software coding standards employed on the software project.

The build utility for embedded software will be the GNU gmake.

The build utility for Windows-based software will be the compiler-provided Integrated Development Environment (IDE), and the compiler-provided Package and Deployment utility.

In addition, other supporting-processes methods, tools, techniques are documented as follow:

- SQAP document 201111208, Section 5 (Standards, Practices & Conventions) and Section 9 (Tools, Techniques & Methodologies).
- SCMP document 201111186, Sections 5.1 (Software Tools), and Section 5.2 Techniques.
- SVVP document 201111254, Section 4.6 (Tools, Techniques & Methodologies).

6.3 Infrastructure Plan

This section of the SPMP specifies the plan for establishing and maintaining the development environment, and the facilities required to conduct the software project.

6.3.1 Office Space

As per Table 5-1, an office space required for this project is for a staff of 7 people. A rearrangement of engineering space is required such that team members can work in proximity.

6.3.2 Personal Computers

The Personal Computers and software needs will be communicated to the MIS department **2 to 4 weeks** ahead of the need-date in order to have the equipment ready in time.

The Table 6-1 summarizes the needs for the personnel depicted in Figure 4-2. The column *software* lists the software required that are in addition to the standard tools deployed at VSRS. All computers will be connected to the corporate LAN.

Table 6-1: Computers and Software Needs

Qty	Group/Facility	Software List
1	Software Management	Microsoft Project SCM Tool
1	Software Engineering – Embedded	Embedded Compiler SCM Tool
1	Software Engineering – Windows-based	Windows-based Compiler SCM Tool
1	Software Quality Assurance	SCM Tool
1	Software Configuration Management	SCM Tool
1	Test Computer for Windows-based and Embedded software	NOTE: Administrator access is required.

6.4 Product Acceptance Plan

The product acceptance plan is documented as part of the Software Verification and Validation Plan (SVVP) document number 201111254. Among other things, the SVVP plans for the development of the Door System's Software-System Acceptance Test Plan, which documents the methods by which the product acceptance will be demonstrated. Note that the SATP should be created and executed by client.

7 SUPPORTING PROCESS PLANS

This section contains reference plans for the supporting processes that span the duration of the software project.

These plans include:

- Configuration Management Plan
- Verification and Validation Plan
- Documentation Plan
- Quality Assurance Plan
- Reviews and Audits Plan
- Problem Resolution Plan
- Subcontractor Management Plans
- Process Improvement Plan

7.1 Configuration Management Plan

Refer to the Door System's Software Configuration Management Plan document 201111186 for the overview of the SCM Functions.

7.2 Verification and Validation Plan

Refer to the Door System's Software Verification and Validation Plan document 201111254.

7.3 Documentation Plan

Refer to the Door System's Software Quality Assurance Plan document 201111208, Section 4 and 5.1 for the list of documentation, applicable standards, and the plan for verification.

Refer to the Door System's Software Configuration Management Plan document 201111186, Section 3.1.1.2 for an exhaustive list of documents and their establishment into project baselines.

7.4 Quality Assurance Plan

Refer to the Door System's Software Quality Assurance Plan document 201111208.

7.5 Reviews and Audits Plan

Refer to the Door System's SQAP document #201111208, Section 6 for Reviews and Audits.

7.6 Problem Resolution Plan

Refer to the Door System's Software Configuration Management Plan document 201111186, Section 3.2 (Configuration Control), and 3.3 (Configuration Status Accounting).

Refer to the Door System's Software Verification and Validation Plan document 201111254, Section 5.5 (Operation), 5.6 (Maintenance), and 7 (V&V Administrative Requirements).

7.7 Subcontractor Management Plans

This section is not applicable to this plan. There is no subcontracted software.

7.8 Process Improvement Plan

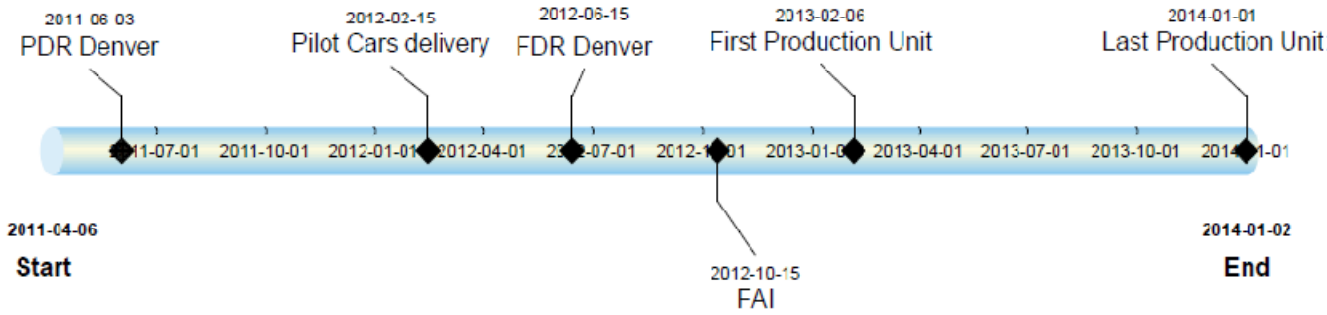
VSRS plans and addresses process improvement at the organization level through continuous process improvement efforts. The Software Process Engineering Group (SEPG) Coordinator coordinates the organization-level process improvements through the guidance of the Engineering Manager.

As the Door System software project matures, the SPMP should undergo revisions reflecting improvements to the processes. Quality and process improvement comes by analyzing and measuring the process in a structured, controlled manner and then changing the process. These improvements should be the product of lessons learned data, findings of audits, and collected process metrics.

8 ADDITIONAL PLANS

To fulfill the safety requirement of the standard EN50128, RAMS engineering group is in charge of producing and maintaining the Reliability, Maintainability, and Safety (RM&S) Program Plan #201111162.

APPENDIX A: SOFTWARE PROJECT SCHEDULES



Note: More precise software project schedules will be produced in the next revision of this SPMP.

APPENDIX B: WORK BREAKDOWN STRUCTURE

Please see the Software Project Development WBS document attached “7230 Denver-WBS-00R02.pdf”